

Student Name: Pierre Thos

Section 1: Tools Declared

OpenAI ChatGPT (GPT-5.5)

Used to explain Otsu's thresholding concepts, simplify mathematical explanations, and help draft study notes in accessible language.

Google Docs AI Grammar Suggestions

Used to refine sentence clarity and improve readability of final written notes.

Section 2: Notable prompts

Prompt 1

"Explain Otsu's thresholding in simple language using analogies."

Response Summary:

The response compared thresholding to separating two hills in a landscape and explained threshold selection as finding the valley between them. This changed our writing approach because it made the concept easier to explain intuitively before introducing formal mathematics.

Prompt 2

"Why does Otsu maximize between-class variance instead of minimizing within-class variance?"

Response Summary:

The response explained the mathematical relationship between total variance, within-class variance, and between-class variance. This helped clarify a concept we initially misunderstood and allowed us to explain the computational advantage properly.

Prompt 3

"Expand this paragraph using journalistic language without sounding overly formal."

Response Summary:

The response rewrote explanatory paragraphs in a more readable style. This improved note quality by making explanations easier to follow while keeping technical accuracy.

Prompt 4

"How do I describe the relationship between within-class and between-class variance mathematically?"

Response Summary:

You can describe the relationship mathematically by treating the image's variance as a constant that is split in two components: between-class variance and within-class variance.

Prompt 5

"Summarize Otsu's original 1979 paper in about 100 words."

Response Summary:

The response provided a concise paper summary that helped structure our key papers section, although we later verified and edited details against the original source.

Section 3: Five questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest thing to explain was how within-class variance and between-class variance work together. At first we understood that Otsu's method tries to make one and the other small. We had trouble saying exactly why it works that way with numbers. The problem was that variance is usually taught in an abstract way so it was hard to see how it applies to image segmentation. We finally got it when we took apart the variance equation and we saw how it balances the two types of variance. This helped us to understand Otsu's method better. It made sense when we related variance to image segmentation in a way; we understood within-class variance and between-class variance properly after that.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

Artificial intelligence initially gave us references that looked like they were from books but they were not complete and we could not find the sources to verify them. One summary that the artificial intelligence generated was also too simple; it did not properly explain why maximizing the difference between classes is easier to do on a computer. This showed us that artificial intelligence can sound very sure of itself when what it is saying is not entirely correct and does not have all the important technical details that artificial intelligence should have.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

We worked with AI to rewrite paragraphs multiple times. The AI kept making the text sound too dramatic or formal. This did not match the tone we needed for our work. We found out that AI works well when you tell it how you want the text to sound from the start. Sometimes it is quicker to edit the text rather than having the AI rewrite it multiple times; the AI rewriting process can be inefficient if not done correctly.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Artificial intelligence made our work go faster when we were trying to rewrite explanations into clearer language for the learning pack. We knew the mathematics behind Otsu's method, however our first drafts were either too formal and hard to follow or too simple to be technically correct. The artificial intelligence helped by rewording these explanations into something more readable while keeping the key technical meaning of the artificial intelligence intact.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

We are still not confident in explaining how Otsu's method deals with images that have severe defects such as for example uneven illumination or histograms with peaks that 'blend' into each other. We also have a hard time with understanding some of the extensions of our topic, such as for example using the two-dimensional Otsu's method.

Student Name : Alexandre Stefel

AI Usage Related to the Adversarial Quiz and Appendix

Section 1 – Tools Declared

ChatGPT (OpenAI, GPT-5.5)

ChatGPT was used throughout the development of the Learning Pack, mainly for brainstorming ideas, improving explanations, refining wording, and generating draft content. For the adversarial quiz, it helped generate misleading distractors based on common misconceptions about Otsu's thresholding. It was also used to improve the clarity of technical explanations and structure the rationale document.

OpenCV Documentation

The OpenCV documentation was used to verify technical concepts related to thresholding methods. I mainly used it to confirm the differences between global thresholding, adaptive thresholding, and Otsu's thresholding before writing the quiz questions and feedback explanations.

Scikit-Image Documentation

Scikit-Image resources were used to better understand histogram distributions and variance optimisation. These sources helped me create questions related to bimodal histograms, noise sensitivity, and threshold selection.

LearnOpenCV

LearnOpenCV articles were consulted to simplify technical explanations and better understand real-world limitations of Otsu's thresholding. This helped me write clearer answer feedback for the Google Forms quiz.

IEEE Paper – Otsu (1979)

The original Otsu paper was referenced to verify the mathematical foundation of the algorithm, particularly the concepts of within-class variance and between-class variance used in several quiz questions.

Section 2 – Notable Prompts

Prompt 1

Prompt:

"Generate adversarial multiple-choice questions about Otsu's thresholding with misleading distractors based on common misconceptions."

Response Summary:

The AI generated several quiz questions focusing on histogram distributions, variance optimisation, and failure cases. Some distractors were too easy at first, so the prompts had to be refined to make the incorrect answers more believable.

Prompt 2

Prompt:

"Explain why Otsu's thresholding fails under uneven illumination."

Response Summary:

The AI explained that Otsu uses a single global threshold and therefore struggles when lighting conditions vary across different image regions. This explanation was later simplified and adapted into quiz feedback.

Prompt 3

Prompt:

“Create feedback explanations for both correct and incorrect answers in a Google Forms adversarial quiz.”

Response Summary:

The AI produced concise feedback explaining why the correct answer was right and why the distractors were plausible. Some explanations initially sounded repetitive, so they were rewritten manually.

Prompt 4**Prompt:**

“Compare adaptive thresholding and Otsu thresholding in simple terms.”

Response Summary:

The response explained that Otsu uses one threshold for the entire image while adaptive thresholding computes thresholds locally. This comparison was useful for designing one of the quiz questions.

Prompt 5**Prompt:**

“Give examples of common student misunderstandings about image thresholding.”

Response Summary:

The AI identified misconceptions such as confusing thresholding with denoising or contrast enhancement. These ideas helped improve the quality of the adversarial distractors.

Section 3 – Reflection Questions**1. Which concept in your topic did you find hardest to explain to someone else, and why?**

The concept I found hardest to explain was the relationship between within-class variance and between-class variance in Otsu’s thresholding. At first, I understood the formula mathematically, but I struggled to explain intuitively why minimising one automatically maximises the other. The explanations often became too theoretical or too statistical, which made them difficult to simplify for beginners. I realised that understanding a concept personally is different from teaching it clearly to someone else. Eventually, visual examples using histograms helped me explain the idea more effectively.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

One issue I encountered was that the AI initially generated quiz questions that were technically correct but too general for the assigned topic. Some questions focused on unrelated computer vision concepts instead of Otsu's thresholding specifically. At first glance, the questions looked convincing, but after reviewing the project brief, I realised they did not fully align with the topic requirements. To fix this, I refined the prompts and manually checked every question against the lecture material and external sources such as OpenCV documentation.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

A moment where AI wasted time was when I asked it to generate "advanced" quiz questions without being specific enough. The output became overly complicated and included concepts beyond the scope of our topic, such as machine learning segmentation methods. I spent time removing unnecessary material and simplifying the questions again. This taught me that AI responses depend heavily on prompt quality. If prompts are too broad, the responses become less useful and may introduce irrelevant information.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

AI was particularly useful when generating plausible distractors for the adversarial quiz. Creating believable wrong answers manually was surprisingly difficult because the distractors needed to sound realistic without being correct. The AI quickly produced several misconceptions commonly associated with thresholding, such as confusing denoising with segmentation or misunderstanding histogram behaviour. This saved a significant amount of time and improved the overall quality of the quiz because the questions became more challenging and aligned with the assignment requirements.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Even after completing the Learning Pack, I still do not feel fully confident explaining how Otsu's thresholding behaves mathematically in very noisy multimodal histograms. I understand the general limitation that Otsu assumes two dominant classes, but predicting exactly how the threshold changes under complex histogram conditions remains difficult. This is because the optimisation process becomes harder to visualise intuitively once multiple peaks and noise distributions overlap.

Student Name: Nathan Soulhol

Assigned Topic: Block 7, Topic 8 — Otsu's Thresholding & Adaptive Thresholding

Section 1: Tools Declared

- **Gemini (by Google):** I utilized Gemini strictly as an interactive study partner and technical assistant to accompany me during my development process. It helped clarify the statistical concepts of Otsu's algorithm, assisted me in diagnosing plotting bugs, and acted as a language partner to polish the English phrasing of the documentation I wrote, as English is not my native language.

Section 2: Notable Prompts

Note: My original interactions with the AI during our live development sessions were conducted in French, my primary working language. For the purpose of this academic report, they are presented directly in English below.

Prompt 1

- **Prompt:** "I want you to read this document and create a document in French explaining what needs to be done and divide the work in 4."
- **Response Summary:** The AI acted as an advisor, summarizing the syllabus requirements and proposing an effective organizational model to split the responsibilities evenly among our team members.

Prompt 2

- **Prompt:** "Explain to me how to build the simulator in more detail step-by-step, and what I should do to be efficient."
- **Response Summary:** The AI provided a structured, step-by-step roadmap for my development process, explaining how to set up the Streamlit interface and structure the local binarization comparison.

Prompt 3

- **Prompt:** "Here is my basic code for the simulator, it is poorly structured, can you restructure it and integrate the sliders for the adaptive method?"
- **Response Summary:** I provided my raw simulator draft, and the AI helped me reorganize it into a clean, functional layout. It resolved my histogram plotting issues and showed me how to map Streamlit's interactive sliders to OpenCV's adaptive thresholding parameters.

Prompt 4

- **Prompt:** "Can you explain this adaptive thresholding code in detail and how its parameters work?"
- **Response Summary:** The AI provided a deep explanation of my adaptive thresholding code

Section 3: Five Reflective Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The most challenging part was explaining the mathematical logic of minimizing within-class variance. I had to use the AI as a sounding board to verify that my simplified explanations remained mathematically accurate.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

The AI provided an incorrect syntax correction that wasted a significant amount of time. I had to independently identify the error introduced by the AI and manually correct the syntax to resolve the issue.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

During the implementation of my histogram code, the AI generated an incomplete code, the AI kept returning slightly different but equally broken lines of code. This taught me that trying to make an AI self-correct syntax errors is highly inefficient compared to checking the official compiler errors and documentation directly.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The AI acted as an invaluable language partner. Because English is not my native language, I spent a lot of time struggling to write technical explanations that sounded natural and professional. By having Gemini I was able to transform my literal translations into clear, academic-grade English. This saved me hours of editing and allowed me to focus my energy on writing the actual Python code and understanding the algorithms.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

The mathematical foundation of the algorithm is highly complex, and while I understand the general concept, some of the deeper equations still remain unclear to me.

Student Name : Eliot Péchadre

AI Usage Related to the Presentation Slides

Section 1: Tools Declared

- **Gamma:** I utilized Gamma strictly as a design and formatting assistant to build the presentation slides. After independently drafting the full content—consisting of a 10-slide outline with 3 bullet points per slide—I used the AI to optimize the visual layout, formatting, and overall presentation design.

Section 2: Notable Prompts *Note: The initial content generation and structure were completed independently. Gamma was used solely to transform this raw text into a polished presentation.*

Prompt: "[Insert your text here / e.g., Convert this 10-slide draft into a professional presentation layout]"

Response Summary: The AI acted as a graphic layout tool, taking my structured bullet points and automatically generating a clean, visually engaging, and professional slide deck.